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# Exploring motivational mechanisms to learn in a national chemistry projects competition: lessons from a retrospective study

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**Abstract**
The "We Have Chemistry!" national projects competition was originally developed in 2008 as a non-formal, engaging learning environment to increase high-school students' motivation to study chemistry. Participants submit chemistry-related projects under STEM/STEAM categories of their choice. A previous study demonstrated that participation might increase students' motivation to learn chemistry. This retrospective study focuses on gaining insights into the benefits of this long-running educational program, elucidating possible underlying mechanisms by which participation impacts students' motivation in relation to competition format and the development of interest. Forty-three competition ex-participants participated in this study, which followed a mixed-methods approach. Causation coding qualitative analysis of narrative data revealed three plausible motivational mechanisms related to the competition format and its components: revealing new contexts for chemistry, engaging in self-directed learning, and targeting personal interest, which were all supported by quantitative data analysis results. Our findings also suggest that for some students, the competition format might facilitate the further development of a personal interest in chemistry that they inherently possess. The combination of contextualized and personalized learning seems to give rise to non-traditional learning processes that target interest, nourish intellectual and emotional needs, and are connected to the motivation to learn chemistry. Results also highlight the teachers' role in promoting student enrollment. The implications of this research apply to the design of other educational programs that aim to increase adolescents' motivation to learn science: they support both the practice of combining non-formal and traditional school settings and the practice of designing learning environments that address student heterogeneity.

**Keywords** Science competitions, Non-formal learning, Motivation, Development of interest, Contextualization, Personalization, Causation coding analysis

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## Introduction

Fulfilling the psychological needs of autonomy, competence, and relatedness was shown to be related to motivation and interest development (Deci & Ryan, 2000; Krapp, 2005; Krapp & Prenzel, 2011). Competitive settings have the potential to foster motivation by allowing their participants to satisfy these basic psychological needs in relation to specific fields of interest. For this to happen, a balance between elements that hinder and foster motivation must be attained, and then the competition can become a scaffold in which nurturing extrinsic and intrinsic motivation might happen when participants exercise related feedback routes (Hong et al., 2019; Ryan & Reeve, 2021). For example, STEM (Science, Technology, Engineering, and Math) competitions might enhance motivation and influence career choices (Höffler et al., 2019; Steegh et al., 2021), and well-designed STEAM (Science, Technology, Engineering, Arts, and Math) settings can support the development of individual interest from situational interest (Hong et al., 2019). With science and STEM competitions gaining popularity (Graffin et al., 2022; Miller et al., 2018; Verhoeff, 1997) and funding (Tschisgale et al., 2024), understanding their role in shaping learning experiences and affecting motivation to learn is crucial.

As part of ongoing efforts in Israel to increase students' motivation to study science and follow STEM careers, the "We Have Chemistry!" national projects competition was launched in 2008 as a joint initiative between the Chemical Industry and the Chemical Academy. In 2013, the Ministry of Education joined the initiative and officially supported participation in it. The program has been running annually, except for two years after the onset of the COVID-19 pandemic and over the first 12 rounds, the number of participants grew gradually from 300 to 1445 per year. An evaluation study of the competition took place in 2011–2012 (Sharaabi-Naor et al., 2014) where changes in motivation were measured for two groups of students who underwent experimental and controlled interventions: students who prepared and submitted projects as part of their participation in the competition and students who prepared and submitted projects under similar conditions but at school. Motivation was assessed in terms of the related dimensions of interest, enjoyment, importance, effort and difficulty, where interest was measured before and after the intervention. Causality was demonstrated between participation in the competition and an increase in the motivation to learn chemistry. However, one of the limitations of the study was that the mechanisms behind the measured increase in motivation were not elucidated. In addition, while the study also demonstrated that the competition positively affected most of its participants in terms of interest, enjoyment, and importance, a decrease in interest was reported for some participants. The main conclusion was that while participation in the competition might increase student motivation to study chemistry, motivational processes are complex and deserve further research.

Regarding the structure of science educational settings of a competitive nature, these can take different forms, such as science fairs, Olympiads, STEM challenges, and science-oriented contests. The impact of these frames on motivation, interest in science, and career choice has been investigated (Miller et al., 2018; Wirt, 2011; Schmidt & Kelter, 2017). Additionally, different aspects of the structure of these frames have also been studied, such as voluntary vs. mandatory participation, competitive vs. non-competitive approach, individual vs. team modes of participation, single vs. multi-round stages, as well as modes of feedback and participant support (Grinnell et al., 2020; Tschisgale et al., 2024; Garrecht et al., 2024). However, to the best of our knowledge, no study has fined-tuned into the different elements of a competition's format and their possible impact on the motivation of its participants to learn science.

We base the present study on our previous research findings and the above-mentioned research gap in the literature. We set out to elucidate the possible relationships between participation in the competition and the motivational processes that contestants undergo when progressing through the competition stages as dictated by its format. By competition format, we refer to the structure that emerges from the organizational components that make up the competition. Specifically, it includes elements such as stages of participation (e.g., enrolling, working on the project, submitting, presenting orally in the final round, etc.), the rules and requirements that participants must fulfil, and the mechanisms of support (mentoring, feedback) provided to students and teachers. We focus on the competition format because its components scaffold the process of participation. We understand the competition format as an important factor shaping the experience of learning in the competition and expect it to affect the choices and actions of participating students. Thus, we assume that different elements in the competition format might act as mediator variables that can directly affect situational interest, which can trigger interest and/or help sustain a well-developed individual interest, also affecting motivation. This assumption is based on interest development models that frame how circumstances affect the triggering of situational interest and how interest is maintained and unfolded into a well-developed interest and the strong link between interest and motivation found by Sharaabi-Naor et al. (2014) and others (Krapp, 2005; Slot et al., 2020). The assumption is further supported by motivation theories that strongly emphasize the situated nature of motivational processes. These theories frame how these processes are driven by

the value and meaning attributed to tasks, expectancies, and self-concepts and that socio-cultural as well as socio-cognitive conditions play an essential role of comparable importance to the role played by internal states, highlighting the integral interplay between the individual and their environment (Eccles & Wigfield, 2020; Ryan & Deci, 2020).

Thus, the analytical structure that we develop to suit our research aims and questions is a combined theoretical framework that addresses the theoretical approaches and related considerations which informed the competition format design, as well as models of interest development in relation to motivation, as we describe in the following section.

## Theoretical background

The competition format was initially developed through research-based design to incorporate two approaches that were previously found to positively affect motivation, interest, and attitudes regarding science, namely, Project-Based Learning (PBL) and Context-Based Learning (CBL). For this study, however, the theoretical framework was expanded to include Personalised Learning (PL) given the high level of choice provision that the competition format allows its participants (regarding both content and learning strategies), as will be described in detail in a following section. In addition, as explained above, we incorporate models of interest development as a suitable lens through which the possible influence of the competition format on motivation to learn chemistry can be conceptualized.

### The PBL, CBL, and PL relationship

PBL is an instructional approach designed to involve students in multi-level problem-solving by emulating real-world scenarios, prompting students to work as a team and prepare projects that are the outcome of self-driven inquiry (Kokotsaki et al., 2016). PBL science competitions are a specific approach to PBL where participants prepare projects that pertain to scientific disciplines, dealing with a STEM subject of their choice and complying with the competition requirements. These frames constitute competitive, non-formal educational environments where participants encounter opportunities for exercising choice and self-directed learning at several stages of their work. In these situations, interest-based learning of science can occur (Young et al., 2017), and it is empowered by decision-making (DeMink-Carthew & Olofson, 2020; Kokotsaki et al., 2016).

PBL is always contextual since it frames the issue around which the projects revolve in real-world applications or circumstances. One feature of PBL-based competitions is that content can be contextualized in a personalized manner when students, working on their projects, choose a scientific topic of their own interest and pose their own questions to drive their inquiry (Schmidt & Kelter, 2017). For example, during a chemistry projects competition, a certain chemistry topic and its related concepts may be explored under different types of contexts (Bennett & Lubben 2006; Habig et al., 2018; Parchmann et al., 2006; Stuckey et al., 2013). One type of context might be the broader socio-cultural-technological-environmental context: where and how chemical concepts are interconnected to the varied aspects of everyday life. The Science and Technology for Society and the Environment (STSE) approach (Gallagher, 1971) falls in the category of such a context. Another type of context might be the chemical context (Gilbert, 2006): each of the chemistry subjects in which certain concepts find representation and meaning. Thus, while working on their projects for a PBL chemistry competition, participants start to localize the chosen topic in different contexts. In this way, context-based learning (CBL), which has been linked to positive effects on both interest and motivation (King, 2012; Menthe & Parchmann, 2015), is inherently present in this kind of PBL-based educational setting, and it can happen guided by students' preferences. In science education, CBL is an instructional method that presents scientific concepts in real-world or relatable scenarios, with the goal of making learning more meaningful and engaging for students and improving understanding (Sevian et al., 2018).

Another feature of PBL-based competitions is the personalization of the learning process, which happens through the provision of choice and the need to make decisions in order to make progress in the project. For example, at the pedagogical level, students can gain knowledge, understanding and skills related to disciplinary (chemical) concepts on a variety of topics and through a variety of strategies, and they can present their outcomes in a variety of different artefacts. Thus, personalized learning (PL) is also inherently present in PBL-based competitions.

In contrast to traditional "one-size-fits-all" educational approaches, PL aims to produce teaching methodologies that address student heterogeneity (Miliband, 2006; Zhang et al., 2020). Although discrepancies exist pertaining to an exact definition and unifying theory of PL, there are agreed principles shown to foster personalised modes of learning. These involve considering and addressing learners' inherent diversity in interests, needs, abilities (Easa & Blonder, 2024), and level of engagement, namely, cognitive and affective factors. Eight basic components of PL have been identified (Walkington & Bernacki, 2020): Individualisation, differentiation, personal adaptation to learning standards, student ownership, the presence of social interaction, flexibility in a learning environment, continuous formative assessment, and

addressing individual interest. The guiding principle nowadays is that providing students with choice facilitates PL (Cheong Li & Tak-Ming Wong, 2020; Zhang et al., 2020; Feldman-Maggor et al., 2023). Different PL strategies award students some degree of control over different aspects of their learning, such as learning objectives, content, context, resources, methodologies, and/or outcomes, all of which can be attained by students learning in a well-structured environment that requires them to make decisions.

At this stage, we can define the relationship between PBL, CBL, and PL. In general, CBL can be brought to students in many formats, such as scientific literacy tasks or short contextualized problem-solving tasks, and it does not necessarily require engaging in PBL. Similarly, implementing PL strategies necessitates neither partaking in the preparation of projects nor engaging in the contextualization of content, and it can be done by granting students choice over certain aspects of the learning process. Conversely, CBL and PL are two intrinsic features of PBL: exploring content through real-world scenarios and engaging in self-directed inquiry and personal decision-making. In parallel, engaging in PBL was found to have a positive impact on motivation, situational interest, and personal interest (Markula & Aksela, 2022; Penuel et al., 2022). Therefore, since it was found that contextualization and personalization of the learning context are positively related to both motivation and personal interest, then non-formal, well-structured PBL science and STEM/STEAM competitions inherently rise as potential platforms for nurturing science-related interest development through CBL and PL and consequently impact the motivation to learn science.

### Models of interest development and motivation

Interest is defined as a personal preference to engage with an object or task (Slot et al., 2020; Bergin, 2016). It is generally categorized between short-lived, externally triggered situational interest and stable, intrinsic, individual interest (Krapp & Prenzel, 2011). Motivation is the process directing behaviour towards a goal (Buck, 2019) and is generally categorized into intrinsic and extrinsic. According to Self-Determination Theory (SDT), intrinsic motivation leads to engaging in tasks because they are personally enjoyable, associated to a personal interest, and support the fulfilment of one or more basic psychological needs (autonomy, competence, and relatedness), while external motivation is mainly driven by external rewards (Ryan & Deci, 2000a). Motivation is regulated by feedback loops, and thus it can also emerge from engaging in tasks which satisfy these psychological needs. Motivation has also been strongly related to the value a person attributes to a task or the expectancies of success pertaining to the performance of a task. The Expectancy-Value Theory (EVT) (Eccles, 2005), for example, proposes that motivation is a product of an individual's expectations for success and their subjective valuing of a task, all of which are shaped by a complex, iterative system. According to EVT, the subjective value a person attributes to a task can be understood in terms of four main mediating factors, namely intrinsic value (sometimes called interest value, defined as "the anticipated enjoyment one expects to gain from doing a task"), attainment value ("the relative personal or identity-related importance attached by individuals" to the performing of a task), utility value (the usefulness attributed to a task, or "how well a task fits into an individual's present and future plans"), and cost value (the net value derived from both the efforts invested and the benefits gained from performing a task). Findings from research pertaining to these and other frameworks consistently show a positive relationship between interest and motivation (Eccles & Wigfield, 2020; Ryan & Deci, 2020), specifically intrinsic motivation (Weber, 2003), and their essential role in promoting learning (Dewey, 1913, 1986; Schiefele, 1991, 2009).

Personal interest has been linked to motivation and defined as a content-specific state bearing intrinsic and value-related aspects (Schiefele, 1991). Dewey (1913) already identified the "activating" nature of interest as well as its personal and contextual characteristics. For example, a decline in motivation to study science was reported in adolescence and has been associated with the structure of the educational context and the lack of student autonomy (Vedder-Weiss & Fortus, 2011, 2012). Therefore, when that's the case, non-formal (Coll et al., 2013) yet well-structured educational settings, like science competitions, could potentially prevent a decline in motivation by allowing the learner to engage with science-related subjects while exerting personal choices regarding the learning process (Verhoeff, 1997) and by targeting personal interest (Chittum et al., 2017).

Interest as a construct and its role in learning have been extensively researched (Harackiewicz & Hulleman, 2010; Hidi, 2006; Renninger & Hidi, 2011, 2019; Schiefele, 1991, 2009). Distinctions are made between situational and individual interests (Slot et al., 2020). Hidi and Renninger's four-phase model proposes that interest develops mainly through four sequential stages: triggered situational interest, maintained situational interest, emerging individual interest, and well-developed individual interest (Hidi & Renninger, 2006). This model emphasises the importance of the learning environment in facilitating the transition between stages.

In their paper on the Situated Expectancy-Value Theory (SEVT) of motivation (Eccles & Wigfield, 2020), an expanded model of EVT, Eccles and Wigfield explicitly drew the connection between the "situational interest"

construct from the four-phase model of interest development (Hidi & Renninger, 2006) and the construct of "intrinsic value" in SEVT, explaining that both focus on liking, enjoyment, and interest in an activity being performed. They also tied the more stable "personal interest" construct from the four-phase model of interest development to the construct of "intrinsic motivation" in SDT, explaining that both are expressed through a deep level of engagement in the activity being performed, through the extent to which individuals want to do the activity, and engage in it independently, regulated by internal rewards. These frameworks all describe the affective engagement that draws individuals toward an activity. Since *intrinsic value* in SEVT is not only about "valuing" a task but about the enjoyment and interest experienced while doing it, which drives intrinsic motivation in SDT (Ryan & Deci, 2000a), interest functions as an affective-cognitive bridge between value attribution (SEVT) and motivation (SDT).

Another model of motivation that closely relates it to the unfolding of personal interest from situational interest was introduced by Csikszentmihalyi (1995) to frame the development of learning in museum settings based on Flow Theory (Beard, 2015; Csikszentmihalyi, 1990). Here, too, the model describes the characteristics of the sequential circumstances that may lead to an increased will to learn, which stems from developed personal interest.

These two models, Csikszentmihalyi's as well as Hidi and Renninger's, share structural parallels and elements. Both models consist of 4 sequential stages, where first, the learner's attention is caught by a trigger or "hook", aimed at targeting the learner's curiosity. Then if the learner has a previous object of interest related to the trigger, or if the task is perceived as personally meaningful, sustained attention gives way to a maintained situational interest. At this stage, the learner must be offered opportunities to engage actively and to interact with the object (which by now can be an existing or a new object of interest). These opportunities are still "external" in the sense that the learning circumstances were previously designed and brought to the learner by the educators, not created by the learner. From this situation onwards, if the task is self-regulated by personal feedback mechanisms, then intrinsic rewards regulate this process; the levels of involvement and difficulty may gradually increase and pose a motivating challenge to the learner. Acquired knowledge and skills are intrinsically rewarding, and partaking in learning satisfies some or all the psychological needs related to intrinsic motivation, as also proposed by SDT (Ryan & Deci, 2000a). In the fourth last stage, if the learner seeks to re-engage with the object of interest (to which he was exposed by the trigger or hook) and thus manifests continuing motivation to learn, it can be concluded that the individual underwent a process through which interest was developed or deepened.

Additionally, numerous studies connect interest and relevance to learning motivation (Frymier & Shulman, 1995; Hulleman & Harackiewicz, 2009; Renninger & Hidi, 2019; Suriakumaran et al., 2017; Stuckey et al., 2013). Other studies relate motivation to learn with supported interest development and engaging learning environments (Krapp, 2005; Ramey et al., 2015). Positive psychology frameworks associate motivation with the assignment of personal meaning (Kaplan et al., 2009; Maehr & Meyer, 1997; Vollstedt & Duchhardt, 2019; Suriakumaran et al., 2017) and recognize that meaning-seeking situations are expected to be both personally interesting and motivating.

The models that frame motivation and the development of interest mentioned above contribute supporting evidence that sequential educational formats that allow students to contextualize content according to personal relevance and provide them with choice could lead to the development of interest and increased motivation. We find that both SDT and SEVT, which account for the situational nature of motivational processes, are suitable frameworks through which to conceptualize the relationship between interest and motivation as seen in this study. Both are inclusive theories of motivation that inherently recognize the interconnection between motivation and interest, as well as the roles of decision-making (choice), perceived usefulness, and relevance as core motivational variables. These are main aspects of the competition format presented here, as will be explained in the next section.

Thus, relying on the combination of the above-mentioned theoretical frameworks and our understanding of its relation to the competition format, we hypothesise that the increased motivation to learn chemistry found in a previous study (Sharaabi-Naor et al., 2014) could be associated with the experience of participation in the competition, and that this relationship is mediated by specific components of the competition format which might be related to motivational and interest-related processes. In the following section, we describe the competition format in more detail and in relation to the frameworks presented here.

### The "We Have Chemistry!" projects competition format

As mentioned in the introduction, the competition format is the structure that emerged from the program's research-based design. It is composed of different rules, requirements, and support mechanisms ordered sequentially according to the competition stages, and it scaffolds the learning experience that the participants undergo while participating in the competition. To participate in the competition, students prepare chemistry-related

projects on which they work for four to five months before submitting them for a first judging round, where projects are judged based on a digital form of the project and written material. Projects that pass this judging round are presented orally at the regional judging round, and if they pass, they proceed to the final judging round, which takes place at the national level in a finalists' conference. Here, the selected projects are presented orally for a second and last time, and the competition winners are announced.

Thus, during the competition, participants progress through up to four main stages: 1) exposure to the program and enrollment (all participants), 2) working on the project and submitting for the first judging round (all participants), 3) working on the presentation and presenting orally during the regional judging round (those who pass), and 4) working on the improved presentation and presenting orally during final judging round (those who pass). Each stage requires that students make decisions and/or contextualize content to fulfil competition requirements. A more detailed description of the competition format, research-based design and its evolution was published elsewhere (Marchak et al., 2023; Sharaabi-Naor et al., 2014; Shwartz et al., 2020) and is out of the scope of this paper. We proceed here to describe the competition format, detailing how PL and CBL are reflected in different stages of the competition.

#### Competition format and the provision of choice

From the beginning, the participants' work is guided by a series of choices that they are required to make. The first of these choices concerns enrollment in the competition. Sometimes, the teacher decides that a whole class will participate as part of an alternative assessment. However, enrollment is freely open to all chemistry students from 9th grade (junior high) to 12th grade (if they major in chemistry in high school), regardless of the teacher's availability or readiness to participate. Students learn about the competition either directly through their teachers, through posters that are sent to schools, and science and chemistry teachers post them on the classroom and lab walls, or by participating in one-day seminars which take place at different points throughout the country and span the north, centre, and south districts. During the first 10 rounds of the competition, chemistry teachers and students attended these seminars (face to face) to hear inspiring chemistry lectures from experts in the chemical industry and academia, as well as partake in workshops according to the category of their choice. In the last two rounds of the competition, and with the aim of making this stage accessible to all, the seminars were replaced with tutorials that can be found online on the competition website (Marchak et al., 2023).

The second and third choices that students make concern the number of group members (although individual participation is also an option), and the type of project (category) that participants want to prepare. There are two major areas from which to choose: either "Science Communication" or "Laboratory Inquiry". Choosing the "Science Communication" area leads to preparing essentially STEAM (Science, Technology, Engineering, Arts, Math) projects where chemistry is integrated with content and skills from another discipline related to the field of communication media, design and/or visual arts. In this area, contestants can choose from 6 available communication formats: poster, photography, short video, scientific article, comics, and podcast. Participants who choose the "Laboratory Inquiry" area engage in self-planned laboratory work, and the product of their work is presented in the form of a laboratory report. This category is intended to emulate the research process in the field of experimental chemistry, and it is only open to 11th and 12th graders who already have a background in inquiry.

Then, participants are required to choose, guided by their curiosity, an "object" on which to work: a chemistry concept, topic, chemical product, or phenomenon about which they would like to know more. In some specific cases, where the teacher asks students to enroll as part of their alternative assessment in chemistry class, participants are somewhat restrained to choose from a list of curricular topics.

After making these first choices, which must be made before enrolling since fields about these issues must be filled out in the digital enrollment form, participants are ready to start working, and it is at this point that they pose self-generated questions to guide their learning.

While working on their projects, contestants must address the competition requirements. They can do so by guiding themselves independently with the sole help of the assessment rubrics and competition guidelines that can be found on the competition website. Alternatively, they can turn to their teacher and/or to the competition staff for guidance and support, according to their needs. In general, chemistry teachers act as facilitators of the learning process and support their participating students to some extent: from exposing them to the project and assisting them in maintaining the competition timetable, to meeting for brainstorming discussions and extra chemistry classes to help them reach the required chemistry level. The competition staff also offers close support to both students and teachers in various ways, ranging from answering questions about technical issues and clarifying assessment rubrics, contacting scientists, industry and science communication experts, to providing feedback about the structure of unfinished projects to help students meet the requirements (before

submission). Regardless of the level of support, participants must make personal decisions as they work on their projects. These decisions can range from advancing personal preferences to agreeing on more technical issues that may arise while working on the products and preparing for the oral presentations. In Table 1 we present the different stages of the competition, describing how each stage might advance personalized learning through the provision of choice.

### Competition format and contextualisation

One of the major requirements of the competition is that a section of the prepared project explicitly acknowledges the influence that chemistry has on the individual at a personal level, on society, and/or on the environment (by a STSE approach). Participants must do so in relation to their object of interest by associating it in a socio-cultural-technological-environmental context of their choice. The goal of this requirement is to make students pay attention to the broader context of the object they chose and make them aware of the existing connections between the chemistry content learned in the classroom and the real world. Another major requirement of the competition is that participants must examine their object of interest from the chemical point of view in connection with the chemistry curriculum by explicitly identifying and addressing the chemistry principles that are related to or can be used to explain the topic/concept/phenomenon/product that they chose to explore. Here, they must refer in their explanations as much as possible to all three levels of understanding of chemical content: the macro, micro, and symbol levels (Johnstone, 1991). Both requirements appear clearly in the guidelines and in the assessment rubric, among other assessment criteria.

These two requirements of exploring the object of interest, in both the STSE and the chemical contexts, serve as guides for learners while they interact with chemistry content. They were specifically introduced for two reasons: 1) to raise students' awareness of the relevance of chemistry at a personal, societal, and environmental level (Hofstein & Kesner, 2006), and 2) to convey academic rigor to participants' work by submitting

**Table 1** Description of the "We Have Chemistry!" competition by its main and sequential stages, which might encourage PL and CBL, as shown in the table

| Competition Stage | Personalization of Learning (by PL) | Contextualization of Content (by CBL) |
|---|---|---|
| Exposure to Program and Enrollment (choices made at this stage must be mentioned in the enrollment form) | Choosing to enroll (although sometimes the teacher's decision as alternative assessment task).<br>Option to participate in a one-day seminar and choosing to partake in workshop with experts according to the intended project category (face to face, first ten years). Choosing to learn about project category through short tutorial videos filmed by experts (online, two last years). | During one-day seminars students and teachers are exposed to lectures from experts in the chemical industry and the academy and learn about applications of chemistry and research in the field (real-world contexts of chemistry content). |
| | Choosing with whom to work (individually or in a group of up to 4 members). | |
| | Choosing project category; to present work outcomes in the form of a Science Communication product (poster, short video, article, podcast, comics, photography) or a laboratory report (following the performance of an experiment). | |
| | Choosing object of interest freely (the chemistry concept, topic, product, or phenomena, about which to learn) or from a list (when teacher decided about enrollment). | |
| Working on the project and submitting for first judging round | Choosing STSE context (situating the object of interest in a personal, social, cultural, technological, industrial, and/or environmental context/s and making the connection explicit). | |
| | Choosing Chemistry context (situating the object of interest in the context of chemistry and provide scientific explanations at least at the level required by curricular standards). | |
| | Making decisions about other issues (driving questions, role division during work, working timetables, information acquisition, information processing, design issues, asking for help from teacher or competition staff or none, working with experts from communication media and/or chemistry fields, etc.) | |
| Working on presentation and presenting orally during regional judging round | Making decisions about presentation-related issues (content, design, roles during presentation, etc.) | |
| Working on presentation and presenting orally during final judging round | Making decisions about presentation improvements and new presentation-related issues (content, design, roles during presentation, etc.) | |

projects that are at least at the level of the chemistry curriculum standards in Israel by connecting their language to the way chemistry is taught in school today, which addresses these three levels of understanding (De Jong et al., 2013).

Opportunities for self-expression and self-directed learning are provided within the educational space delineated by these two requirements, together with the degrees of choice that students encounter at several stages of their work (see Table 1). These degrees of choice are reflected in the variety of projects that were presented over the years. For example, one group of participants can decide to investigate nail polish as a chemical beauty product in the context of the stabilisation of pigment solutions in evaporable solvents, compare them to similar solutions that exist in the car industry, and then present their findings as a scientific article. Another group may investigate the intermolecular forces between the nail polish components and the nail, explain the side effects that these products can have on nail health, and then present their project as a short movie. A third group can approach the same beauty product, nail polish, in the context of polymer chemistry, give a historical and chemical account of its evolution over time, and then present their work through a well-designed scientific poster. These three groups all chose "nail polish" as the object of interest but situated it in different STSE and chemistry contexts, as well as chose different means of communication to express their findings. These are examples of the variety of outcomes that result from undergoing consecutive stages of decision-making and contextualization throughout the competition, which is unavoidable if contestants want to fulfil all its basic requirements. Table 1 summarizes the competition format, including the main sequential stages and shows how participants are provided with choice and required to contextualize content at every stage.

### Research aims and questions

As explained in the Introduction, this study aims to better understand the nature of the motivational processes allowed by the competition (Sharaabi-Naor et al., 2014), examining the possible relationships by which participation might affect students' motivation to learn chemistry, focusing specifically on gaining insight regarding *how* the competition might benefit its participants. Our main hypothesis is that an increased motivation to learn could be related to certain aspects of the competition format, and we assume that there should be components in the competition structure that would directly impact the learning experience that participants undergo, probably supporting the nourishment of a newly developed or an existing interest in chemistry. We therefore posed the following research questions:

(1) What were the motives that encouraged students to participate in the competition?
(2) What were the students' perceived gains from having participated in the competition?
(3) What are the mechanisms behind the competition's positive influence on the motivation to learn chemistry?

## Methods

### Population

We recruited former high-school chemistry students who had participated in the competition in previous years and who had already finished high school. Recruitment was done by e-mail, after receiving IRB approval, by sending letters to all ex-participants who were older than 18 years of age at the time of sending the e-mail and for whom we had an e-mail address in the competition enrollment system (from 12 years of participation records). From about 800 e-mail letters that were sent and not automatically rejected by e-mail systems, 43 ex-participants voluntarily agreed to participate in this research. We note here that we do not know how many of these 800 e-mail addresses were effectively active (in use) at the time of sending the letters.

We defined this $N = 43$ sample as "ex-participants" to emphasise the retrospective nature of the data gathered for this study; we requested them to think about the experience of partaking in the competition in a retrospective manner. We decided to focus on this population to gather information about how participation was perceived by reflecting on it from a broader, more mature point of view. According to the hypothetical model of causal relations (Schiefele, 1991), such an account is regarded as a "post-actional" account of an experience. In comparison to an account of an experience during the "actional" phase (such as the one that could be produced while reflecting on an experience that is ongoing), which is expected to rely mainly on emotional and cognitive processes, a "post-actional" reflective account is expected to integrate in addition an appraisal of learning outcomes as well as an attribution of personal meaning and value to the whole process.

Statistical characterization of the sample revealed that it is representative of the general distribution of the competition participants' population diversity in terms of gender, religion, and demographics. For a full list of the survey questions as they were asked in the questionnaire (Q1 to Q14) and statistical characterization results, see Supplementary Information 1. Since, in this study, we focus mainly on potential benefits from having participated in the competition and seek to explore ex-participants perspectives about how the competition might increase the motivation to learn chemistry (if they think it does), we regard this population as a purposeful

**Table 2** Likert-type questions as asked in the questionnaire

| | |
|---|---|
| Att1 | Learning chemistry in school was useful. |
| Att2 | Learning chemistry in school was not interesting (negative). |
| Att3 | Learning chemistry in school was relevant to me. |
| Att4 | Learning chemistry for the project was useful. |
| Att5 | Learning chemistry for the project was not interesting (negative). |
| Att6 | Learning chemistry for the project was relevant to me. |
| Att7 | Participation in the competition made me think differently about chemistry. |
| Att8 | Participating in the competition showed me aspects of chemistry that I didn't know before. |
| Att9 | Participation in the competition made me more interested in chemistry. |
| Att10 | In the competition, I learned things that were not just about chemistry. |
| Att11 | Participation in the competition did not personally contribute to my knowledge of chemistry (negative). |
| Att12 | I am glad I participated in the competition. |

To what extent do you agree with each of the following statements? (scale of 5-points answers: (5) largely/(4) to some extent/(3) little/(2) very little/(1) not at all)

sample and understand that most ex-participants that have decided to voluntarily answer the research questionnaire might have a positive outlook of the competition.

### Research tools and data analysis

Volunteering ex-participants filled in an anonymous digital questionnaire designed to supply quantitative as well as qualitative data. It included survey questions, 12 Likert-type questions, and three open-ended questions, as described below. The resulting data were analysed using a mixed-methods approach (Creswell, 2014): 1) Survey data were statistically examined to characterize the research population. 2) Likert-type items underwent descriptive statistics to determine how ex-participants rated certain aspects of the experience of learning chemistry in the competition and learning chemistry in school for further comparison. 3) Qualitative data obtained from open-ended questions pertaining to the drives and gains concerning participation in the competition were subjected to non-hierarchical, flat coding analysis, categorisation, and summative frequency counting to reveal the motives behind participation and the perceived rewards from having participated. 4) Another set of qualitative data obtained from open-ended questions underwent causation coding analysis (Saldaña, 2013) to elucidate plausible motivational mechanisms by which participation in the competition increases the motivation to learn.

The 12 Likert-type questions (Table 2) consisted of two sets designed for different purposes. The first set (Att1 to Att6) was designed to compare how respondents graded learning chemistry in the competition and in the classroom by three specific, well-known motivational variables: usefulness, interest, and relevance (Vollstedt & Duchhardt, 2019). This comparison was meant to quantify ex-participants' perspectives about both educational settings in basic motivational terms. The second set (Att7 to Att12) was designed to assess certain aspects related to the participation process as understood from a retrospective point of view and to quantify the overall experience of having participated. Reliability analysis of the Likert-type scale was performed using DATAtab software by calculating Cronbach's alpha for set 1 (0.84) and for set 2 (0.83). We also calculated Cronbach's alpha for the sub-set Att1-Att3, which refers to learning chemistry in school (0.69), and for sub-set Att4-Att6, which refers to learning chemistry for the project (0.73), verifying that items in these sub-sets are consistent in their measurement to an acceptable extent.

**Table 3** Open-ended questions as asked in the digital questionnaire

| | |
|---|---|
| OQ1 | What motivated you to participate in the competition? |
| OQ2 | Would you recommend chemistry students to participate in the competition? If so, why? |
| OQ3 | In your opinion, does participating in the competition encourage the study of chemistry? If so, how? |

Additionally, we analysed open-ended questions OQ1 to OQ3 (see Table 3). Qualitative data collected through these questions were subjected to different types of narrative analysis.

OQ1 and OQ2 were designed to help clarify the motives behind the decision to participate in the competition (expected gains), as well as the possible components of participants' satisfaction regarding participation in the competition (realized gains) accordingly. While students' reasons to participate are assumed to be influenced by personal preferences regarding chemistry and motivational factors existing before the competition, the reasons for recommending participation are assumed to be influenced by the experience of having participated. By examining the gaps between expected and realized gains, we can better understand the motivational factors that were probably elicited during participation. Data obtained from answers to these questions were examined by non-hierarchical (flat) coding, categorization, and frequency analysis (Saldaña, 2013). At all stages, coding, labelling, and categorisation were validated by other researchers in the group. After a preliminary round of raw data analysis by the first author, the two main researchers examined the raw data, discussing, agreeing and/or modifying the flat coding, labelling, and categorisation choices. In the following rounds of analysis, these results were discussed with the second and third authors to examine the adequacy of coding and, consequently, fine-tune labelling and adjust categorisation. The discussions between the researchers continued until full agreement was reached. To establish the reliability of the data

analysis, the resulting interpretation of the data was discussed independently with an additional researcher, who is not one of the authors. During the meeting, the overall analysis of the data proposed by the main researchers was corroborated, supporting the confidence in the validity and robustness of the interpretation.

The full analysis process is described in detail as follows: we began by carefully reading the written accounts, searching for words or pieces of text that could be tagged as a key concept/s used to answer the question. These concepts were encoded into labels that were assigned as a basic paraphrasing, matching a definition of the code they represent (flat coding labels for questions OQ1 and OQ2 appear in Supplementary Information 2). All labels were assigned the same weight and then decoded into categories that arose from the analysis: each label was designated as belonging to one of these emerging categories. Following the theoretical framework, the attribution of the emerging categories was done by integrating motivation literature (Deci & Ryan, 2000; Hunter & Csikszentmihalyi, 2003; Ramey et al., 2015) with findings from neuro-psychological research regarding typical teenage behaviour (Blakemore, 2008, 2012; Blakemore & Robbins, 2012; Schweizer et al., 2019) and findings from research regarding personal meaning-making in youths and adolescents (Bar-Tur et al., 2001; Fave et al., 2013; Luciana & Collins, 2012; Ratner et al., 2019; Schnell, 2011), all of which support the affective, cognitive and social aspects of the model of interest development (Hidi & Renninger, 2006). Accordingly, categories were expressed in terms of affective, intellectual, social, and sensation-seeking motives for participating (in OQ1) and reasons for recommending participation (in OQ2). The last step in the qualitative analysis of the answers to these two questions involved summing up the frequencies of the labels to calculate the weight of the emerging categories for each of these questions.

OQ3 (Table 3) was introduced to encourage ex-participants to produce personal verbal accounts related to the mechanisms that might have led to an increased motivation to study chemistry. This question requested ex-participants to explain how participating in the competition could encourage the study of chemistry if they thought it did. To answer, respondents should evaluate the connection between the experience of participation and wanting to re-engage with the study of chemistry, assess it from a post-actional point of view (Schiefele, 1991), pinpoint the competition's attributions, and construct a causal explanation. Accordingly, we applied causation coding (Saldaña, 2013), identifying in each text the following components: 1) *the cause*: Code1, 2) *the outcome*: Code3, and 3) *the relationship* (link between cause and outcome): Code2. In most cases, causality could be identified in the relationship, giving rise to the sub-set Code2A (cause of Code2) and Code2B (outcome of Code2). The process is summarized by the following term, where the symbol "→" means "leads to":

$$\text{Code1} \rightarrow \text{Code2} \rightarrow \text{Code3} = \text{Code1} \rightarrow \text{Code2A} + \text{Code2B} \rightarrow \text{Code3}$$

In OQ3, Code1 and Code3 were posed by the question itself: *The cause*, Code1, being "Participation in the competition …", and *the outcome*, Code3, being "… thus encouraging the study of chemistry" or equivalent variations of these expressions. For clarity, a given answer to this question is analysed in the following example: "*Yes, the competition reveals more sides of the world of chemistry and materials, which pushes the students to be interested in the field of chemistry.*" In this example, the question was answered affirmatively, accepting that Code1 leads to Code3. In addition, causality can be identified in the relationship so that Code2A is tagged as *"reveals more sides of the world of chemistry and materials"* and Code2B as"*pushes the students to be interested in the field of chemistry*".

**Table 4** Mean and median values obtained from Likert-type questions Att1 to Att6

| | Variable Short | Mean | Std Dev. | Median |
|---|---|---|---|---|
| Att1 | School Useful | 4.79 | 0.71 | 5 |
| Att2 | School Interesting | 4.28 | 1.12 | 5 |
| Att3 | School Relevant | 4.30 | 1.01 | 5 |
| Att4 | Proj. Useful | 4.14 | 1.04 | 4 |
| Att5 | Proj. Interesting | 4.30 | 1.08 | 5 |
| Att6 | Proj. Relevant | 4.00 | 1.09 | 4 |

*N* = 43

## Results and their interpretation

### *Likert-type variables analysis*

Results from descriptive statistical analysis of Likert-type questions Att1 to Att6 are presented in Table 4. The mean values and standard deviations were calculated after corroborating that the distribution symmetry for each variable was adequate. Median values were calculated since they provide valuable information regarding the tendencies of the data set (Norman, 2010; Sullivan & Artino, 2013). The mean values of the scores from the first set of questions resulted in averages above 4 for all items, the median values being either 4 or 5. In the frame of a 5-point scale, these values indicate that at least 50% of the participants rated the variables highly, suggesting that students who participated in the competition equally appreciated both chemistry learning frames (school and competition) in terms of usefulness, interest, and relevance. The only statistically significant difference between the average scores, according to the t-test, was the 13.57% higher mean score attributed to the usefulness

of learning chemistry in high school (Att1) in relation to learning chemistry for the project (Att4) ($p < 0.0001$).

From the mean values of the scores that ex-participants gave to the second set of Likert-type questions, Att7 to Att12 (Table 5), we can see that all averages are above 3.7, and the median values fall between 4 and 5. This suggests that most ex-participants underwent a process that usually involved a positive change in the way they think about chemistry, in what they know about chemistry, an increased interest in chemistry, the acquisition of new general skills and knowledge, and increased chemistry knowledge. There is an overall satisfaction from having participated in the competition, which is evident from the high score that item Att12 got, namely, 4.53 with a deviation of 0.83.

Of the 43 ex-participants in our sample, 12 did not choose to participate in the competition. These explicitly mentioned that enrollment was compulsory as part of the class requirements. Observing the data pertaining to this specific group showed that, although they were obliged to participate, most of them (10/12) still gave overall high scores to the processes they underwent while participating in the competition (Att7 to Att11). Within this group, 10 out of the 12 rated "being glad to have participated" (Att12) between 4 and 5 on the scale, and the remaining 2 rated this item as 1 and 3.

During the interpretation of these and all results, we kept in mind that data were collected from subjects who voluntarily answered the questionnaire and thus might have had a positive attitude towards the competition. We elaborate further on this point in the following section.

Quantitative analysis results indicate that for most ex-participants in this study, learning chemistry in the competition was perceived as useful, relevant, and interesting. This could be understood in terms of the value possibly assigned to high-school chemistry studies when evaluated retrospectively regarding the matriculation diploma, admission to universities, and schemes of future plans. Since usefulness, relevance, and interest are accepted motivational values (Vollstedt & Duchhardt, 2019; Suriakumaran et al., 2017), these results suggest that for the studied sample, both frames were perceived as motivating chemistry learning frames. In addition, most ex-participants benefited from participating and were glad to have participated ($N = 41/43$). Based on the principles of Situated Expectancy-Value Theory (SEVT) as detailed in the Theoretical Background section (Eccles & Wigfield, 2020), the findings from the quantitative analysis can be further interpreted in terms of the subjective value that ex-participants attributed to participating in the competition. The finding that most ex-participants perceived the competition as "useful" during a retrospective recollection of it can be seen as related to an attributed utility value and suggests that having participated might fit in with their academic goals or academic self-concept. Regarding relevance, finding the participation in the competition relevant can be interpreted as a manifestation of attainment value, which is related to how a task or activity confirms or expresses central aspects of one's identity. By participating and finding the experience relevant, ex-participants may have been affirming an aspect of their own identity as competent students, or a "science person," or someone capable of meeting an academic challenge of this sort. The perception that the competition was "interesting" corresponds directly to intrinsic value (or interest value). This component points to an inherent enjoyment that ex-participants might have experienced by participating in the competition, suggesting that it might have allowed the generation of this type of value.

As will be shown below, the interpretation of these results is supported by the answers to the hypothetical recommendation to participate in the competition (OQ2), where 35 out of 43 answered affirmatively, including 11 out of 12 within the "no choice" group. We inferred from these results that the researched sample underwent an overall positive experience by participating in the competition, apparently regardless the choice to enroll.

At this stage, it is worth mentioning that we did look for differences between sub-groups: by year of participation, by gender, by type of category, and by choice to participate. However, we found none, and we attribute this to the small sample size, which, upon division, leads to no statistically significant differences.

**Table 5** Mean and median values obtained from Likert-type questions Att7 to Att12

| | Variable Short | Mean | Std. Dev. | Median |
|---|---|---|---|---|
| Att7 | Change in the way of thinking about chemistry | 3.70 | 1.32 | 4 |
| Att8 | Discovering new aspects of chemistry | 3.91 | 1.29 | 4 |
| Att9 | Increased interest in chemistry | 3.81 | 1.28 | 4 |
| Att10 | Learning new things in general | 3.88 | 1.22 | 4 |
| Att11 | Increased knowledge in chemistry | 3.91 | 1.21 | 4 |
| Att12 | Glad about participating | 4.53 | 0.83 | 5 |

$N = 43$

### Flat coding analysis

Qualitative data obtained from answers to open-ended OQ1 and OQ2 were examined by flat coding, as explained above, resulting in counted frequencies for each label and weights for each category. Flat coding is beneficial in exploratory analysis to prevent the loss of information and to refrain from interpreting data during the first stages of analysis before defined structures arise. The full list of the labels that emerged from the process of

encoding (including definitions) can be found in Supplementary Information 2, as well as a listing of the categories that emerged during the decoding process and their assigned definitions.

In OQ1 we asked ex-participants what motivated them to enroll in the competition. Thirty-three out of the 43 ex-participants answered this question. The following is an example of one of the given answers: *"My chemistry teacher actually managed to convince me. On the one hand, there was the fact that winning the competition could exempt you from the matriculation oral examination in the laboratory, and on the other hand, research and chemistry in general are things that I really like, so that every opportunity I had to expand my horizons on the subject, I went for it and the competition was part of that"*. In this example, we identified the following labels: the teacher, winning (sensation seeking), love of chemistry, and extra-curricular chemistry (wanting to learn beyond the content learnt in school). Table 6 shows the categories that emerged from OQ1 analysis and the labels that belong to each category. These categories and their labels can be understood as the causes that motivated students to participate in the competition; thus, we defined them as "drives". It is worth noting here that the categories of "No Choice" and "The Teacher" differ. "No Choice" pertains to those students who did not choose to enroll but were obliged to do so ($N = 12$, 16 being the frequency of no-choice-related verbal terms that were counted in their answers), whereas "The Teacher" category pertains to those students who mentioned having been encouraged by the teacher (but not obliged to enroll): *"My teacher proposed it to me, and it sounded very interesting, so I decided to participate"*.

**Table 6** Flat coding results from analysis of OQ1: what motivated you to participate in the competition? $N = 33$

| Label | Label Frequency | Category (and Summative Frequency) |
|---|---|---|
| Personal preference | 4 | Affective Drive (6) |
| Wish to excel | 1 | |
| Personal progress | 1 | |
| Love of chemistry | 10 | Chemistry-related Affective Drive (10) |
| Curiosity | 3 | Intellectual Drive (21) |
| Personal interest | 10 | |
| Learning | 3 | |
| Researching | 3 | |
| Extra-curricular chemistry | 2 | |
| Teaching program | 12 | No Choice (16) |
| Compulsory | 4 | |
| Group work | 1 | Peer Interaction (4) |
| Senior students' role model | 1 | |
| Social interest | 2 | |
| Visiting the Weizmann Institute of Science | 1 | Sensation Seeking (7) |
| Challenge | 2 | |
| Out-of-school | 2 | |
| Winning | 1 | |
| Fun | 1 | |
| The teacher | 11 | The Teacher (11) |

In OQ2, ex-participants were asked whether they would recommend chemistry students to participate in the competition and, if so, why. Thirty-five out of the 43 ex-participants answered positively, including 11 out of the 12 ex-participants who were obliged to enroll in the competition. All 35 who answered positively provided an explanation for their answer, such as the following exemplary student statement: *"Yes, I would recommend participating in the competition because while preparing the project, I learned in depth about the topic that I chose, and thus you can learn and understand a certain topic thoroughly. Also, participating in the competition exposes chemistry students to different and diverse topics from the world of chemistry and allows them to learn about them from other students, so each student learns and receives an explanation about topics other than the topic he researched. In addition, participating in the competition provides the experience of the lecture, the transmission of an explanation in front of a group, strengthening such skills in the student."* In this statement we could identify the following labels: tailored learning (control over choosing the topic), a platform for deepening knowledge, extra-curricular chemistry (learning beyond the content learned in school), social aspects of learning, tagging learning as "an experience", and public speaking. Table 7 shows the categories that emerged from analysing the answers to OQ2 and the labels that belong to each category. These categories and their labels can be understood as the reasons behind recommending participation in the competition; thus, we defined them as "gains".

By comparing the reasons to enroll in the competition with the reasons for recommending participation (summarized graphically in Fig. 1a and b), we can learn about the difference between the anticipated and the actualised rewards from having participated in the competition.

We found that from all the self-reported reasons for enrolling in the competition, intellectual drive and chemistry-related affective drive were the most significant self-motivating factors. By enrolling in the competition, ex-participants were willing to invest the efforts involved in participating in the competition, expecting to benefit at the intellectual, affective, and social levels for different specific reasons, such as a willingness to learn or research, fulfilling a curiosity or a personal interest

**Table 7** Flat coding results from the analysis of OQ2: would you recommend chemistry students to participate in the competition? If so, why? $N = 35$

| Label | Label Frequency | Category (and Summative Frequency) |
|---|---|---|
| Recognition | 1 | Affective Gain (6) |
| Ownership | 1 | |
| Sense of success | 1 | |
| Personal connection to a scientist | 1 | |
| Personal connection to chemistry | 1 | |
| Emotionally worthwhile | 1 | |
| Everyday aspects of chemistry | 5 | Intellectual Gain (45) |
| Personal interest | 8 | |
| Platform for research | 6 | |
| Extra-curricular chemistry | 15 | |
| Platform for deepening knowledge | 4 | |
| Creative thinking | 1 | |
| Analytical skills | 1 | |
| Broader chemistry knowledge | 2 | |
| Interesting experience (in relation to learning) | 3 | |
| Learning differently | 4 | Non-Conventional Learning (21) |
| Learning experience | 4 | |
| Peer learning | 2 | |
| "Tailor-made" learning | 6 | |
| Interview a scientist | 1 | |
| Platform for creative expression | 4 | |
| Social aspects | 5 | Peer Interaction (8) |
| Public speaking | 2 | |
| Collaboration | 1 | |
| Independence | 1 | Personal Growth (6) |
| Personal choices | 2 | |
| Personal progress | 1 | |
| Perception change | 1 | |
| An empowering experience | 1 | |
| Winning | 1 | Sensation Seeking (34) |
| Platform for experiencing | 5 | |
| Breaks routine | 7 | |
| Out-of-school | 1 | |
| Fun | 10 | |
| "An experience" | 10 | |

related to chemistry, or wanting to engage in the extra-curricular learning of chemistry. Additionally, ex-participants often mentioned that they enrolled because of their love of chemistry. From these findings, we infer that most participants in this study appear to have an inherent personal preference for chemistry, which existed before the competition to some extent. These data also revealed that the encouragement of the teacher to enroll, non-obligingly, was an important external motivating factor.

Regarding the reasons for recommending participation in the competition, the most frequent self-reported reasons pertained to intellectual gain, sensation-seeking gain, and non-conventional learning (Fig. 1b), which we understand as realised gains from having participated. By partaking in the competition, ex-participants seemed to have realised an inherent intellectual preference for chemistry and were emotionally rewarded. In addition, they frequently reported engaging in learning chemistry in non-conventional ways as a reason for recommending participation. Based on these results, we interpret that the competition format fostered the nourishment of an intellectual need in addition to being perceived as enjoyable and satisfying, and it is thus worthy of being recommended. These findings, together with the overall high score given to the statement "being glad to have participated" (Att12, Table 5) points to a positive perceived cost value that emerged from having participated in the competition, which, according to SEVT, is understood as the net difference between the invested efforts and the realized expectations from the efforts (Eccles & Wigfield, 2020).

### Causation coding analysis

Causation coding was performed on the answers to OQ3: "In your opinion, does participating in the competition encourage the study of chemistry? If so, how?" Of the 43 ex-participants, four ex-participants answered "not necessarily", and two of them followed to elaborate: *"It seems to me that there is no connection between the two. If someone is already studying chemistry and participating in a competition, it is interesting and exciting, but it will not make people want to learn chemistry from scratch."* and *"From what I understand, someone who participates in the competition is already learning chemistry."* Two ex-participants who did not choose to participate answered that participation in the competition does not encourage you to study chemistry and stated that: *"Not at all, because the teacher forced everyone to choose a specific topic and did not let them choose the topics [that] the students wanted."* and *"No, it does not encourage studying chemistry because they require [putting] emphasis on simple subjects that are taught in school."* Eleven ex-participants did not address this question.

Ultimately, 26 ex-participants answered affirmatively to OQ3, of whom 24 followed to elaborate and explain *how* participation encourages studying chemistry. Some accounts ($N = 10/24$) highlighted the importance of uncovering new contexts of chemistry as a motivational actuator: *"Yes. Participating in the competition shows aspects of chemistry that are not usually exposed in regular matriculation studies and can therefore help students who do not connect to formal studies, in that the competition shows them how much chemistry is in each field of*

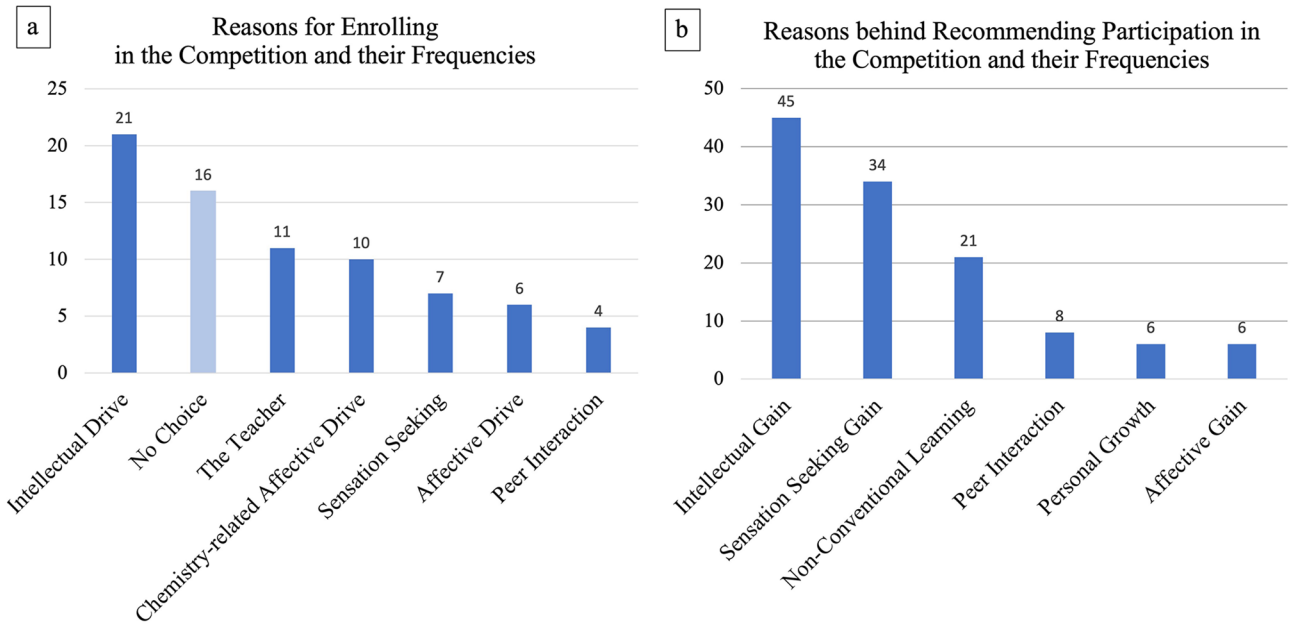


**Fig. 1** A graphical summary of the categories resulting from flat coding analysis and the summative frequencies of the labels associated with each category. (**a**) Question OQ1 - the emerging reasons behind enrolling in the competition ($N=33$); (**b**) Question OQ2 - the emerging reasons behind recommending participation in the competition ($N=35$)

*life."* or "*Yes, the competition reveals more aspects of the world of chemistry and materials, which pushes students to develop an interest in chemistry."* Some ex-participants ($N=6/24$) proposed explanations related to the opportunity to engage actively and independently in the process of learning chemistry: *"Very much. Students usually learn in a lecturing method. The competition included self-research and choosing a completely independent topic, you prepare a project with a lot of freedom. You see in projects the touch of each student and the reason why he chose this project."* or "*Yes, [you] learn in an interactive and unusual way, students teach you different topics*". Others ($N=8/24$) wrote about how the competition aroused curiosity, or encouraged them to target an existing interest: *"Yes, it arouses interest by using the content learned in class to learn a specific topic that interests the student."* or *"Yes, it creates curiosity and interest and a desire to promote the research done in the process, and thus creates motivation to learn in the field that the students chose"*.

Results from analysis of OQ3 are summarized in Table 8, including short excerpts from the accounts shown according to each of the three mechanisms to which they were assigned (numbered 1, 2, and 3 in Table 8). Identifying an explanation as belonging to a certain mechanism was done according to the attribution expressed by ex-participants in the first part of the textual account (Code2), where the connection between the cause (Code1 - "Participation in the competition …") and the main reason that gives rise to the relationship (sub-set Code2A) was clear. The attribution expressed in the second part of the textual account (sub-set Code2B), if present, could be identified as an elaboration of the relationship, supporting it rather than leading to the outcome (Code3 - "… encouraging the study of chemistry"). The summary in Table 8 shows that 1) for some students, seeing the place of chemistry in everyday life can lead to various outcomes such as increased interest, enjoyment or connecting chemistry with other fields; 2) for others, engaging in active, independent learning can lead to various outcomes such as learning being more fun or experiencing a positive impact on interest; and 3) for other students, an experienced increased curiosity or interest in chemistry can lead to discovering the relevance of chemistry in every-day life, or being encouraged to inquire on subjects about which they became curious. These are understood as distinct motivational mechanisms that causally connect participation in the competition with an increased motivation to study chemistry.

The three emerging motivational mechanisms summarized in Table 8 are interpreted here in terms of the competition format and the theoretical framework developed for this study, particularly through the lens of SDT (Deci & Ryan, 2000) and SEVT (Eccles & Wigfield, 2020), as they were described in the Theoretical Background section:

1) "Participation in the competition reveals chemistry in new contexts". We interpret this mechanism as being related to the requirements regarding the contextualisation of content by both STSE and chemistry contexts. Participants are required to explicitly show at least one

**Table 8** Causation coding results from the analysis of OQ3. $N = 24$

| | | | | |
|---|---|---|---|---|
| **Participation in the competition …** | **reveals chemistry in new contexts (mechanism 1)** | | | **… thus, encouraging the study of chemistry.** |
| | uncovers, shows, helps discover ($N = 10$) | applications of chemistry in any field | | |
| | | the chemistry behind the products in our lives | connecting to other fields | |
| | | another side of chemistry | being very enjoyable | |
| | | the many fields in chemistry | looking into the eyes of research | |
| | | the chemistry in everything | leading to interest | |
| | | the world of chemistry | making you interested | |
| | | different faces of chemistry | increasing your desire to understand chemistry | |
| | | different aspects of chemistry | helping students who don't connect | |
| | | the taste of chemistry | | |
| | **promotes active, independent learning (mechanism 2)** | | | |
| | fosters, allows, enables ($N = 6$) | to ask questions | being more interested | |
| | | to learn and expand on a selected topic | knowing how to learn differently | |
| | | to learn in an interactive, different way | making studies more fun | |
| | | to learn from other students | researching independently | |
| | | to independently select a topic and research | preparing a project with a lot of freedom and personal touch | |
| | | to learn subjects that you don't learn in class | discovering how much you love inquiring and chemistry | |
| | **targets personal interest (mechanism 3)** | | | |
| | arouses, creates, encourages ($N = 8$) | curiosity | causing motivation to learn | |
| | | interest | showing how the world of chemistry is diverse | |
| | | desire to know more | letting you find motivation to inquire | |
| | | desire to explore | generating enthusiasm | |

of the broader contexts in which the chosen chemistry topic can be placed (personal, socio-cultural, technological and/or environmental), as well as explore the chemistry related to the object of interest to produce scientific explanations.

Contextualisation of content under a real-world context frame might lead to increased motivation by allowing various processes to take place and support the generation of related values (Eccles & Wigfield, 2020), such as the realisation of personally important connections to chemistry content (attainment value), increasing awareness of the relevance and usefulness of chemistry (utility and cost values), fulfilling personal inherent interests in chemistry (attainment value), and allowing new interests in chemistry to emerge by expanding its connectedness to other subjects and everyday scenarios (intrinsic value). Regarding the competition as "revealing of yet unknown aspects of chemistry" emerged as a gain also through the answers to OQ2 pertaining to the recommendation of participation to other students. For example, *"Yes, [it is] an opportunity to discover different and interesting fields related to chemistry, not just the standard things you learn in high school."* From this example of a reply, we can infer that a combination of utility, attainment, and intrinsic values might have been generated from participating in the competition through the discovery of new contexts which are also seen as personally interesting, as well as cost value by a non-standard learning experience, which overall lead to the perception of the participation as a net gain.

This mechanism is also supported by findings from the quantitative data analysis indicating that some ex-participants discovered new aspects of chemistry because of having participated in the competition, as well as having learned new things in general. In terms of SDT, where competence is one of the three psychological needs identified as essential in motivational processes (Ryan Decy 2000a), the requirement of contextualizing content in this manner, shaped by the contextual scaffold around which participants prepare their projects, might affect motivation by supporting the fulfilment of related psychological needs, such as experiencing intellectual competence in chemistry.

2) "Participation in the competition promotes active, independent learning". We associate this mechanism with the high degree of choice and the levels of control that participants have over the learning process while preparing the projects. This interpretation is supported by the finding that the third most significant perceived gain from having participated in the competition was engaging in "non-conventional learning" (Fig. 1b). This gain was not found to have a parallel motive for enrolling, yet it was found to be a significant gain. We attribute this gain to the personalization aspects of the competition format since, while the anticipated rewards can be connected mainly to individual interests and perspectives that existed before the competition, actualised rewards can possess, in addition, interest components and perspectives elicited by the circumstances, namely, situational interest (Slot et al., 2020). In addition to answers

to OQ3 and their analysis (Table 8), we found that some ex-participants' answers to OQ2 further strengthen our interpretation regarding this motivational mechanism: *"Yes, because it's fun to learn about something you choose and create your own product",* or *"Yes, I recommend it, because the competition is an opportunity to learn chemistry in a fun and different way from school (there are also prizes ☺)".*

Apparently, some students can translate the provision of choice into a personal adaptation of the learning process to fit individual learning profiles. As a motivational mechanism, personalization through choice might allow participants to experience autonomy and competence in relation to the process of learning chemistry, which could motivate those for whom "intellectual gain" was a significant gain from having participated in the competition. Thus, requiring personalizing content and the learning process might support the fulfilment of autonomy and competence, both psychological needs identified by SDT as essential in motivational processes (Ryan & Deci, 2000a). In terms of SEVT, the provision of choice was found to assist the subjective attribution of the different components of value to a task through iterative, individually adapted processes (Wigfield et al., 2020).

3) "Participation in the competition targets a personal interest". We interpret this mechanism as being grounded in the requirement of contextualisation of content in combination with the provision of choice for students who already had an interest in chemistry at the time of entering the competition: guided by personal preferences, participants must scientifically explain a chosen topic or phenomenon based on accurate chemistry content. This interpretation is supported by findings from quantitative data analysis and qualitative flat coding analysis, which together suggest that some ex-participants in this study had an inherent interest in chemistry. Many ex-participants found that learning chemistry in school was comparable to learning chemistry for the project in terms of interest, and intellectual gain was the most frequent reason to recommend participation. Here, too, some ex-participants' answers to OQ2 support this interpretation: *"Yes [I would recommend]. For those interested in the field, this is an instructive and interesting experience that takes away from the school routine,"* or *"I would recommend it to anyone who is interested in chemistry and research because this is an opportunity to get information and experience on how research works with a supervisor, how to plan an experiment in a scientific way and how to present the findings in front of an audience."*

Thus, the competition might serve as a platform to further develop or deepen an existing interest in chemistry. As a motivational pathway, the requirement of connecting the chosen topic of the project (a personal object of interest) with chemistry (as suggested, a subject of interest) can lead to a deeper knowledge of chemistry acquired through finding answers to self-posed questions, which were elicited according to what is personally meaningful for the learner. By fulfilling the requirement of producing in-depth chemistry explanations about the chosen topic, participants might have an opportunity to exercise chemistry-related intellectual competence, probably cultivate a pre-existing inclination to engage in chemistry and make personal connections between chemistry and an object that is of interest to the participants because it is important for them at an identity-related level. Targeting an existing interest might lead to the generation of related attainment value (Eccles & Wigfield, 2020) and is understood as an accepted way to increase motivation to learn by many motivational theories (Ryan & Deci, 2000b; Csikszentmihalyi, 1993; Dewey, 1913)

Additionally, given that participants typically work in teams and seek support from the teacher, the competition likely cultivates a sense of relatedness. Our findings that "the encouragement of the teacher" and "peer interactions" were both emerging drives to enroll in the competition (Fig. 1a), and that "peer interactions" was one of the reasons for recommending participating in the competition (Fig. 1b), support this inference. Relatedness is one of the psychological needs that, if fulfilled, can lead to increased intrinsic motivation (Deci & Ryan, 2000), and it can result in a more positive learning experience in terms of affect. Eccles and Wigfield (2020) have also postulated the influence that socio-cultural contexts have on individuals and how, by shaping the circumstances in which they find themselves, they impact a person's beliefs and values and thus the motivation to engage or not in a task.

### Limitations and future research

A key limitation is the relatively small sample size, $N = 43$ out of the 800 emails sent without return, resulting in an under-representative sample and potential information loss. In addition, only 60% of participants ($N = 26$) provided full narrative accounts for OQ3, limiting the analysis scope of motivational mechanisms.

In addition, evaluating the competition retrospectively emerged as both a strength and a limitation. Based on the hypothetical model of causal relations (Schiefele, 1991), the positive side of reviewing an evoked experience is that the post-actional evaluation would include cognitive and emotional outcomes and personal significances attributed to a specific experience in relation to other experiences up to the moment of answering. However, the answers to the questions were elaborated long after having experienced the competition. Since it was shown that emotions play a strong role in long-term

memory and consolidation (Li et al., 2020; Smith-Lovin et al., 1995), we acknowledge that volunteering ex-participants might be emotionally biased, positively or negatively (Karabenick et al., 2007), by how the experience's outcomes and significance are perceived today. One of the reasons for applying a mixed methods approach in this research was to overcome, at least partly, this limitation.

Another limitation is the lack of information regarding how far participants progressed in the competition. While all respondents submitted projects, their exact stages (finalists, winners, etc.) are unknown, which could influence some of the perceived gains as well as their overall experience from having participated.

To address these limitations and better understand the competition's overall impact, a continuing research study is being designed to specifically address participants who are partaking in the competition nowadays and who do not submit projects, as well as non-winning finalists. This will allow us to evaluate broader aspects of the program, including the teacher's role in this non-formal educational environment, its relationship to motivation to learn (Fortus & Touitou, 2021) and project success, as well as the impact of the teacher's incentive versus the teacher's decision to participate on attitudes about the competition and the development of interest in science (Grinnell et al., 2021). Regarding the provision of choice, it was shown that sometimes it might have negative effects when students get lost in decision-making and fail to self-regulate, which can lead to disengagement and frustration (Patall et al., 2008; Schmidt & Kelter, 2017). In addition, the scaffolding of students' decision-making process must also be evaluated since guidance and instruction were found to be of key importance to lead to positive outcomes (Garrecht et al., 2020). Consequently, to complement this study, future research will investigate possible negative consequences (Maoz & Rosenfeld, 2005), lack of motivation or the development of uninterest, as well as long-term effects (Staus et al., 2021) that might result from participation in the competition. Once a comprehensive understanding is formed, evidence-based guidelines on how to foster the positive and minimize the negative aspects of these competitions can be delineated.

Given the limitations mentioned here, we refrain from generalizing our results; we take them as evidence of the beneficial aspects of participating in this long-running educational program and as a guide to further research. Moreover, we emphasize the need for future studies with larger and more diverse samples to robustly test the hypotheses and compare the effects of different competition formats.

## Discussion

The "We Have Chemistry!" national projects competition was originally designed as a non-formal competitive setting to engage students in chemistry-related CBL through PBL with a high degree of decision-making (PL). Our main hypothesis in this study was that an increased motivation to learn could be related to certain aspects of the competition format.

We learned that most ex-participants perceived learning chemistry in the competition as equally interesting and relevant as learning chemistry in school. Regarding usefulness, ex-participants' overall rate for learning chemistry in school was statistically significantly higher than learning chemistry for the competition project. We attribute this difference to the specific retrospective appreciation of curricular chemistry studies from the perspective of an educational-system bachelor, where utility and cost values are possibly assigned to high-school chemistry's significance in relation to matriculation qualifications, university admissions, and future career planning. This interpretation is based on studies that examined how formal science education is valued in terms of real-world applicability, the development of various skills, and integration into the workforce (Lavi et al., 2021; Osborne et al., 2003). We found no relevant literature that has studied the perceived usefulness of participating in a science competition in comparison to high-school science in a population comparable to the one examined by the sample in this study. Thus, this arises as a potential point for further research. In addition, we would like to stress that although subgroup analyses by gender, category, and year of participation did not yield statistically significant differences, we regard this lack of differences as resulting from the small sample size in this study and not as an indicator that there are no differences between sub-groups. Further research with larger samples could address this aspect as well.

Regarding our first research question (What were the motives that encouraged students to participate in the competition?), we found that intellectual drive, the encouragement of the teacher, chemistry-related affective drive, sensation seeking, affective drive, and peer interaction were the reported reasons behind participating in the competition. However, intellectual and chemistry-related affective drives emerged as the most significant intrinsic factors motivating their enrollment. This finding aligns with findings from previous studies (Parchmann et al., 2006; Dierks et al., 2014; Höffler et al., 2019) suggesting that some science competition participants seem to have an inherent inclination for science and that both cognitive and affective aspects of learning are important to develop an interest further and lead to motivation in learning (Buck, 2019, Hannula et al., 2019; Menthe & Parchmann, 2015; Neher-Asylbekov & Wagner, 2023).

Regarding our second research question (What were the students' perceived gains from having participated in the competition?), we found that intellectual gain, sensation seeking gain, non-conventional learning, peer interaction, personal growth and affective gain were the reported reasons behind recommending other students to participate in the competition, which we interpreted as actualized rewards from having participated.

Comparing anticipated and actualized rewards revealed that participants sought to engage with chemistry at intellectual and affective levels. Participation seems to have met students' intellectual and affective needs pertaining to engaging with chemistry, in addition to being perceived as an enjoyable and enriching experience involving non-conventional, self-directed learning. Similar rewards from participating in science competitions were reported in a study by Abernathy and Vineyard (2001), where "learning something new" emerged as one of the significant perceived rewards. Sensation-seeking rewards, such as enjoying the process, having fun, and breaking routines, also emerged as significant. Most ex-participants found the competition worthy of being recommended to other students.

We conceptualised an existing interest in chemistry as preceding the motivation to learn (Harackiewicz & Hulleman, 2010; Hidi, 2006; Tuan et al., 2005). On the other hand, we acknowledge the situated nature of interest development and its relation to motivation and understand that certain situations might support the development of interest and, in turn, increase motivation. Thus, we set out to specifically explore possible relationships between participation in the competition, the competition format, interest development in chemistry, and motivational processes to learn through our third research question (What are the mechanisms behind the competition's positive influence on the motivation to learn chemistry?).

Three motivational mechanisms emerged (Table 8): discovering new contexts for chemistry, engaging in self-directed learning, and targeting a personal interest. These mechanisms are supported by similar findings from science/STEM/STEAM competitions, Olympiads, and science fairs (Young et al., 2017; Dionne et al., 2012). Specifically, Dionne et al. (2012) found five sources of motivation for participating in science fairs: (1) interest in science, (2) sense of self-efficacy, (3) assurance of achievement through rewards or gratifications, (4) social aspects, and (5) strategies to gain scientific knowledge and methods. These sources of motivation overlap with the motivational mechanisms that we propose and together support evidence in favour of an increased or developed interest being facilitated by the competition format.

Because of the interest-motivation relationship, when inspecting the competition format in terms of sequential models of interest development (Bergin, 2016; Csikszentmihalyi, 1995; Hidi & Renninger, 2006; Krapp & Prenzel, 2011) we suggest that the competition format possibly facilitates transitioning from situational interest into a maintained individual interest. Furthermore, increasing the level of interest might lead to an increased motivation to learn (Ryan & Deci, 2000b; Csikszentmihalyi, 1993; Dewey, 1913). However, heterogeneity regarding the participants' initial level of interest in chemistry and the high level of choice provided prevent us from analyzing sequential phases of interest development within the competition structure based on our data in the present study. Our findings provide evidence that some participants might experience a development of interest in chemistry because of participating in the competition and that motivational mechanisms concerning personal interest, contextualisation, and personalised learning might be related to this positive effect. Yet, elucidating the actual process of interest development and transition between different phases would involve further research designed to tackle this question specifically.

As to the beneficial aspects of this long-running, non-formal competitive setting, this study shows evidence that the competition format might enable some of its participants to contextualize content through their work, to experience personalisation of the learning process, and experience the development or deepening of a personal interest towards chemistry; these processes seem to be positively related to an increase in motivation to study chemistry. During the competition, students learn chemistry through well-structured PBL that asks them to contextualize content and engage in decision-making at all stages. To fulfil the requirements posed by the competition in its sequential stages, students rehearse self-preferred learning strategies and practise control over a broad range of components of the learning process; they decide on the learning pace, invested effort, challenge level, content resources, chemistry context, STSE context, modes of learning, levels of creative elaboration of content, levels of collaboration, and self-expression, between other choices. Our present findings show evidence that some participants experience a development of interest in chemistry as a result of participating in the competition and that motivational mechanisms concerning personal interest, contextualisation of content, and personalised learning are related to the positive effect that participation has on the motivation to learn chemistry. These findings are supported by other studies which show that contextualization, personalization, and choice as educational strategies can potentially enhance intrinsic motivation by allowing the assignation of personal meaning to learning (Cordova & Lepper, 1996).

## Conclusions

One of the main findings of this study suggests that certain components of the competition might mediate the increased motivation to learn chemistry that arose from participating in the competition for some participants. The three motivational mechanisms that emerged from pertinent qualitative data analysis were defined as: revealing new contexts for chemistry, engaging in self-directed learning, and targeting personal interest, which were all supported by quantitative data analysis results. Since these findings are based on self-reports, causality cannot be firmly established from the current data. Thus, we refrain from overgeneralizing our results and consider them evidential of the beneficial aspects of participating in this long-running educational program. Through this study, we gained insight into the beneficial aspects of the competition and how its research-based format potentially leads to an increase in motivation to study chemistry. Overall, the implications of this research advocate for educational practices that address students' individual interests, choice, and real-world connections. Thus, our results have possible implications regarding the design of engaging interdisciplinary science learning environments.

The OECD "Knowledge for 2030" concept note on the future of education and skills emphasises the importance of developing procedural and epistemic knowledge interdependently with skills, values, literacies, and agency (OECD, 2019) in parallel to content knowledge in order to give rise to a "new normal" in formal education. An innovative science learning environment can be designed to complement formal science education by incorporating components of non-formal educational settings into formal educational settings while retaining and expanding the main learning goals to include additional learning goals. For example, by allowing students to rehearse several skills (creative thinking, critical thinking, communication, and collaboration), gain science and visual literacies' skills (through contextualization and the preparation of projects that communicate science visually), address learner heterogenicity and support the inclusion of active learning modes (through personalization of the learning process), and help develop and deepen an interest in STEM disciplines. As an example of combining non-formal and traditional school settings, since 2014 Singapore's Ministry of Education has complemented its formal STEM curriculum with a non-formal STEM approach, namely, learning through authentic experiences, providing students with open-ended projects based on their own interests, incorporating science competitions and the involving relevant industry stakeholders; through the country's Science Center Singapore. Research has shown that this STEM Applied Learning Program, which reaches over 60% of the country's primary and secondary schools, helps develop student interest in science and science identity (Dhanaraj et al., 2023; Toh et al., 2021).

Thus, we propose that this non-formal, well-structured science competition setup or an adapted version be considered when designing innovative learning environments in chemistry and other STEM disciplines. We argue for the introduction of open-ended projects based on student interests and authentic contexts that implement content learnt in class, to support student motivation to study science.

## Supplementary Information

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Supplementary Material 1

Supplementary Material 2

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**Author contributions**

All authors contributed to this study. DM, MK, and RB were involved in the study design and in data analysis validation. DM's work also included data collection, data analysis and plotting, and writing the first version of the manuscript. DM, MK, SR, and RB contributed to iterative discussions, results interpretation, conceptualization, reviewing, and editing. RB supervised the work.

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**Data availability**

The datasets used and/or analysed during the current study are available from the corresponding author upon reasonable request. Qualitative data was analysed in Hebrew and was not translated into English.

## Declarations

**Ethics approval and consent to participate**

This study was conducted following IRB research approval.

**Originality and submission to other outlets**

This manuscript reports on original, unpublished research that was not simultaneously submitted to another publication outlet.

**Consent for publication**

All individuals identifiable in this manuscript have given their consent for publication.

**Competing interests**

The authors have no competing interests to declare that are relevant to the content of this article. The authors have no financial or proprietary interests in any material discussed in this article.

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